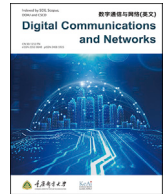




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Dynamic defenses in cyber security: Techniques, methods and challenges

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ABSTRACT

Driven by the rapid development of the Internet of Things, cloud computing and other emerging technologies, the connotation of cyberspace is constantly expanding and becoming the fifth dimension of human activities. However, security problems in cyberspace are becoming serious, and traditional defense measures (e.g., firewall, intrusion detection systems, and security audits) often fall into a passive situation of being prone to attacks and difficult to take effect when responding to new types of network attacks with a higher and higher degree of coordination and intelligence. By constructing and implementing the diverse strategy of dynamic transformation, the configuration characteristics of systems are constantly changing, and the probability of vulnerability exposure is increasing. Therefore, the difficulty and cost of attack are increasing, which provides new ideas for reversing the asymmetric situation of defense and attack in cyberspace. Nonetheless, few related works systematically introduce dynamic defense mechanisms for cyber security. The related concepts and development strategies of dynamic defense are rarely analyzed and summarized. To bridge this gap, we conduct a comprehensive and concrete survey of recent research efforts on dynamic defense in cyber security. Specifically, we firstly introduce basic concepts and define dynamic defense in cyber security. Next, we review the architectures, enabling techniques and methods for moving target defense and mimic defense. This is followed by taxonomically summarizing the implementation and evaluation of dynamic defense. Finally, we discuss some open challenges and opportunities for dynamic defense in cyber security.

1. Introduction

With the continuous development of the Internet of Things (IoT), cloud computing and other emerging technologies, various Cyber-Physical Systems (CPS) have been established in all walks of life, in which information resources are fully shared and utilized concurrently. On the one hand, these resources have become the key strategic infrastructures of all countries and organizations, which support the effective operation of national power, transportation, finance, energy and other important and influential fields. On the other hand, these resources profoundly affect and change people's way of production and life, giving birth to a new normal of social operations [1,2]. Nonetheless, benefiting from the enriching information resources and services, security threats of global cyberspace are also taking on new dimensions. Various cyber security incidents frequently occur while diverse novel cyber-threats are spreading globally. Major security incidents (e.g., Wanna Cry ransomware virus, eBay data breach) have repeatedly shown that cyber security faces serious challenges over the years [3].

In view of defense for cyber security, researchers have conducted extensive findings. The traditional cyber defense technologies (e.g., authentication, access control, information encryption, intrusion detection system, vulnerability scanning and virus protection) have provided a certain degree of security [4,5], whereas with the development of diversification attacks, the traditional cyber defense is inadequate. The existing defense mechanisms are inadequate to prevent various types of attacks, and the dominating reasons include:

1. **The universality of vulnerability.** Limited by the technological capabilities and engineering skills, it is impossible to fully avoid, detect and eliminate vulnerabilities in static hardware/software components, systems, tools, environments and protocols.
2. **The easy installation of backdoors.** Under the globalization of the information industry, it is easy to implant backdoors through the product design chain, the tool chain, manufacturing chain, processing chain, supply chain, service chain, and other links.

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3. **The oneness of genes in cyberspace architecture.** Cyberspace technologies and system architectures have homogeneity (e.g., use the same processor, operating system, office software and database). Due to their static, deterministic and similar situational mechanisms (e.g., system configuration, operation agreement, topology and transport routes), the ecological environment is very fragile. It not only causes vulnerability and makes the backdoor be attacked easily, but also enables the attack chain to be sustained and effective for a long time.
4. **The asymmetry between offense and defense.** From the perspective of attackers, all it takes is a single exploitable vulnerability in the entire security chain to disrupt or take control of the entire system. Meanwhile, it has a target space that is almost free from any constraint. Moreover, they have the initiative to launch sudden attacks at any time. From the perspective of defenders, they have to defend against known and unknown threats in all aspects of the communication network and information system.

Therefore, cyber-attacks based on unknown system vulnerabilities and backdoors are still the greatest threat in communication networks. The inevitability of vulnerabilities and the limitations of perceived defense methods force administrators to change defense strategies and innovate defense mechanisms, so as to reverse the passive situation of being prone to attacks and difficult to take effect in cyber security. Dynamic defense in cyber security based on mobile target defense and mimicry defense rises in response to the proper time and conditions.

Moving Target Defense (MTD) is a game-changer for cyber security proposed by the United States of America (U.S.A.) in view of the current inferior position of the defender [6,7]. It is expected to confuse the attackers by continuous and dynamic changes, so as to increase the cost, complexity and failure rate of the attack [8,9]. It is important to note that MTD is not a specific defense method but a design guideline. MTD does not attempt to establish a system without loopholes, but to employ the resources, time and space environment of the target system to present the attacker with a constantly changing attack surface, which increases the difficulty of the attacker's cognition of the target system and reduces the duration of system vulnerability exposure [10–13]. Therefore, attackers barely develop effective attack methods against the target system in a limited time to improve the resilience and active defense capability of the target system.

Mimic Defense (MD), as a neoteric active defense technology in cyberspace, aims to improve the anti-attack capability of information devices through endogenous mechanisms of its construction. The core idea of MD is to organize multiple redundant heterogeneous functionalities to jointly handle the same external request [14–16]. Meanwhile, MD implements dynamic scheduling based on negative feedback among

multiple redundancies to compensate for the security flaw in the current cyberspace.

In recent years, dynamic defenses of cyber security based on MTD and MD have been frequently investigated in academia and industry. Dynamic defense technologies applied to information systems have been put forward and achieved certain defense abilities. However, research on dynamic defense technologies is still in its infancy at present, and the theoretical study and engineering applications are facing several problems and challenges, such as the theoretical model of dynamic defense mechanism, the mechanism strategy of dynamic defense, the theoretical method of measuring the effectiveness of dynamic defense, and the index system of the influence of dynamic defense on system performance, etc. Therefore, in-depth theoretical study and system improvement of dynamic defense have important theoretical guidance and practical significance for promoting active defense capability.

Although numerous researches and practices on the dynamic defense in cyber security have emerged, there are only a handful of publications that systematically introduce this kind of work. The related concepts and development strategies of dynamic defense are rarely analyzed and summarized. To bridge this gap, a comprehensive and concrete survey of the recent research efforts on dynamic defense in cyber security are conducted in this paper.

The paper is organized as follows. Section 2 introduces an overview of the basic concepts and definitions of dynamic defense in cyber security. Furthermore, Section 3 surveys the architectures, enabling techniques, and methods for MTD in cyber security. Section 4 presents the architectures, enabling techniques, and methods for MD in cyber security. After that, Section 5 reviews the implementation and evaluation of dynamic defense in cyber security. Finally, Section 6 discusses future directions and open challenges of dynamic defense in cyber security.

2. Moving target defense

Moving target defense provides a new way of thinking to solve the problem. At present, a large number of studies have been proposed which involve many aspects of MTD. In this section, we systematically introduce, classify and summarize the existing achievements in MTD. An example of an MTD model is given in Fig. 1.

2.1. Basic concepts

In this section, we introduce two basic concepts in MTD, i.e., attack surface and attack surface conversion. In fact, the notion of attack surface was proposed long before the concept of mobile target defense appeared, which was mainly used as an important indicator to measure system security in the early stage of software development [17]. After the

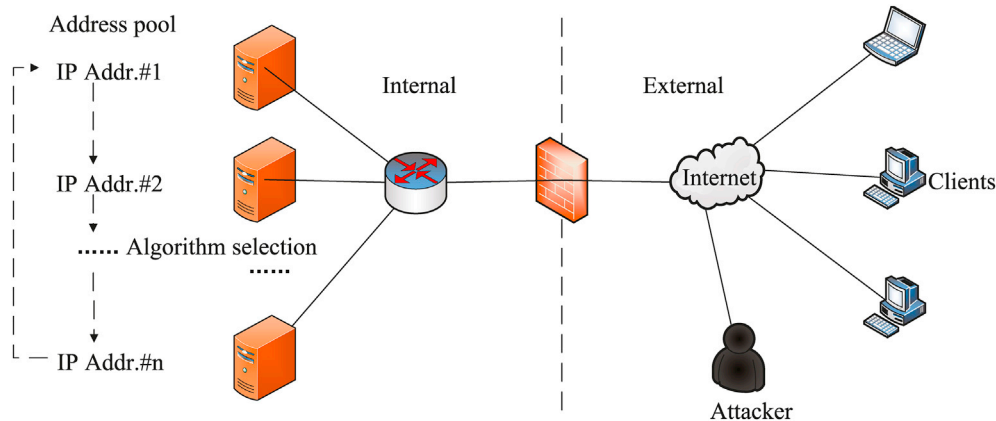


Fig. 1. An example of MTD model.

concept of MTD was proposed, the conversion of attack surface is regarded as an important way to realize moving target defense [18], and some researchers have tried to find a general method of attack surface transformation by using game theory [19,20] or attack graph theory [21] to provide the optimal moving target defense.

2.1.1. Attack surface and attack surface conversion

As a matter of fact, there is currently no standard definition of attack surface [22], and the existing definition is usually relevant to the scenario. Manadhata et al. [23] regarded the system attack surface as a subset of resources utilized by attackers to carry out attacks in the system. Zhuang et al. [24] believed that the attack surface in the system consists of the resources revealed to the attacker (e.g., software on the host, communication ports among hosts and vulnerability points of each component) and network resources that have been compromised and be utilized to enter the system. Zhu et al. [20] regarded the attack surface as the set of vulnerabilities explicit to the system that an attacker might use for the attack. Peng et al. [25] consider the attack surface of an instance virtual machine instance in a cloud service as the total resources available.

Although the concept of attack surface has been widely used in the research of mobile target defense, the existing definition of attack surface still lacks comprehensiveness, accuracy and popularity. Therefore, to better illustrate the defense process against moving targets, it is necessary to further describe the characteristics of the attack surface.

Huang et al. [26] graphically described the transformation process of the attack surface but did not provide a formal definition. After that, Manadhata [19] firstly proposed the concept of attack surface shifting and defined it as follows:

- **Definition 1.** Attack surface parameters. The attack surface parameter represents the system configuration vulnerability or property of the attacker that initiates the attack, including software and hardware configuration property vulnerability of the system, such as buffer overflow vulnerability. In addition, it also includes the network properties exploited by the attacker, such as IP address, service port, and so on.
- **Definition 2.** The attack surface. At any time, the attack surface of the system is determined by the attack surface parameter set and the

specific value of each parameter in the set. The system attack surface at time t is denoted $As = \{M_t, E_t\}$, where $M_t = \{m_{1t}, m_{2t}, \dots, m_{lt}\}$ represents the attack surface parameter set at time t , and $m_{it} (1 < i < L)$ refers to a specific attack surface parameter at time t , whose range is u_i . In addition, $E_t = \{e_{1t}, e_{2t}, \dots, e_{lt}\}$, where $e_{it} \in u_i$ represents the specific value of the parameter $m_{it} (1 < i < L)$ at time t .

- **Definition 3.** For a specific system G , the previous attack surface of G is denoted as R_o , and the new attack surface is denoted as R_n . If there is a resource r that satisfies one of the following two conditions, then the attack surface of G has been transformed from R_o to R_n :

1. r is a member of R_o but not of R_n ;
2. r is a member of both R_o and R_n , but the role of r in R_o is greater than that in R_n .

This definition considers that the transformation of the attack surface can be realized either by changing system resources or by changing the role of a system resource, and it is not easy to quantify the role of resources in the attack surface.

The basic definitions of MTD are summarized in Table 1.

2.1.2. Basic theory

In the MTD system theory [22], a large number of basic definitions of MTD are introduced. To realize the general processing flow of an MTD system, three key problems must be solved: 1) Configuration selection problem. How to choose a new configuration for the mobile target defense system to make it more difficult for attackers to attack the system; 2) Behavior selection. How to choose adaptive behavior to realize the new configuration; 3) Timing choice, which is a key factor that will affect the defense effect and system performance. In Ref. [27], Zhuang et al. define the concepts to facilitate precise discussion of the attacker's knowledge, the types and instances of attack. Moreover, the authors propose some of the design of MTD guidelines and the basic framework of network MTD to improve the resilience of the system under attack, which can be integrated with SDNA and another MTD mechanism [28]. Hobson et al. [29] believed in order to achieve effective movement, three types of challenges need to be solved: 1) Coverage, that is, the movable part in the attack surface, which can be simply defined as the proportion of the dynamic part in the whole attack surface; 2) Unpredictability related to the design of mobile and the range of an attacker to guess or predict the likelihood of mobile implementation, which indicates the attacker's mastery of ground movement information; 3) Timeliness, a move should begin before the end of the attack. Besides, they discussed other considerations for designing and deploying mobile target technology, including low overhead, direct cost, utility, defender experience, and MTD's dependency on other system components.

Carvalho et al. [30] described the background of MTD and some important fundamental problems, such as the premise and threats to build resilience for MTD and to realize moving ahead. Moreover, the authors also analyzed the need for a control and command mechanism that implements system movement logic and provides adaptive responses to failures and attacks [31]. In addition, the authors also raised a control framework based on human-agent team collaboration to facilitate the actual deployment of MTD [32].

Torrieri et al. [33] discussed the problems and challenges needed to study endogenous and exogenous interference attacks and other attacks, and proposed a basic framework to solve such problems based on the concept of cyber maneuver. The whole framework is centered on mobile keys, which can supplement higher-level network keys and provide methods to deal with internal and external attacks. However, the design of each component of the framework and the seamless connection between components have not been solved, so further in-depth study is still needed.

Crosby et al. [34] argued that the implementation of MTD mobile mechanism fully considered the network dependence, and attackers need to rely on specific information to launch successful attacks and proposed that the design of mobile target defense mechanism should start from

Table 1
Summary of the basic definitions of MTD.

Category	Reference	Contribution
The definition of the attack surface	[23]	Resources (e.g., methods, channels, data, etc.) that are utilized by the attacker to launch an attack on a subset of system resources
	[24]	Resources (such as software, ports, etc.) that are exposed to an attacker, as well as network resources that have been compromised and can be used to access the system
	[20]	An explicit set of vulnerabilities of a system that can be used by an attacker
	[25]	A virtual server pool with diversity is taken as an example to illustrate the means of the attack surface movement graphically
The definition of attack surface transformation	[26]	The concept of the attack surface transformation is defined graphically and formally, in which the contribution of resources to the attack surface is very important
	[19]	The transformation of the attack surface is defined graphically and formally, and the main contribution of this paper is the importance of resources to attack the surface

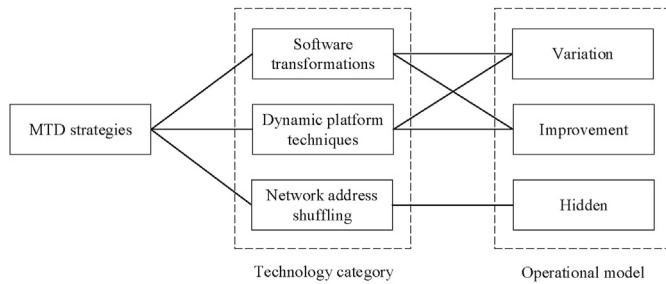


Fig. 2. The technology category corresponds to the operational model.

three aspects, identifying the attackers' dependence on network protocols, services and applications; identifying those dependencies that are broken to confuse, delay, or hinder the attacker and designing corresponding mechanisms to reduce the impact of interrupted dependencies on legitimate users.

Green et al. [35] identified and defined three characteristics of network-based MTD mechanisms: 1) Movement characteristics, including the unpredictability of movement, the vastness of target space, and periodicity; 2) Access control features, including uniqueness, availability and revocability; 3) Recognizability, that is, to distinguish the trusted user from the non-credit account. To verify the correctness of their work, the authors analyzed four NMTD systems (DNS Capabilities Approach, OpenFlow Mutation, MT6D, and Simulation-based MTD) and specified which of the seven characteristics.

2.2. MTD strategy

In this section, we mainly summarize the existing MTD strategies. The strategy of MTD mainly involves three technologies, i.e., software transformation, dynamic platform and network attribute transformation [36]. We give an overview of MTD strategies in Fig. 2.

2.2.1. MTD strategy based on software transformation technology

The mechanism is mainly based on software transform software applications for mobile parameter changes. By using different modifications, different variants having the same behavior and characteristics are used alternately, and there is an uncertain and unpredictable situation in front of the attacker. It is difficult for attackers to smoothly carry out their malicious behavior and increase the difficulty of the corresponding attack by an attacker, improving the ability of software against attacks. This part mainly focuses on the following aspects.

Proactive obfuscation [37] counters an attack by creating multiple copies of a server and periodically restarting a new copy. These multiple copies are generated by semantically preserved code transformations that provide the same functionality with minimal vulnerability in common. That is, they are diverse. The periodic restart policy limits the number of copies of the service that are compromised at any given time, and because two different copies have so few vulnerabilities in common, an attack on one copy is difficult to migrate to the other, effectively increasing the difficulty of the attack but introducing additional copy creation and management overhead.

Pappas et al. [38] proposed a practical software diversification technique, in-place code randomization, to help third-party applications resist Return-Oriented Programming (ROP) attacks. To protect each instance of a binary executable code snippet of code randomization, different transformations are randomly selected and applied, such as automatic sequential replacement, instruction reorganization, register reallocation, etc. A small scale to destroy the code of semantics makes it impossible that an attacker can always find unmodified gadgets for effective ROP attacks at any time. This technique can randomly select and apply different transformations to each instance of a third-party application without changing the location of the basic program block, thus

effectively increasing the threshold of the attacker, reducing the probability of successful attacks, and introducing less additional overhead.

Symbiotic Embedded Method (SEM) [39] is a kind of new coexistence defense mechanism that can be used to protect the device drivers, the kernel and user applications. The working process of this mechanism includes three stages: creating SEM, using existing technology to transform and randomize SEM and protected program, and injecting SEM into the protected program. When an SEM is created and ready to be injected into the program, the SEM and the program which is protected will be analyzed and transformed to create a single instance of the original code. In this process, some existing technologies are used, such as the Aggregation Services Router(ASR), Integrated Services Router(ISR) and polymorphic deformation to increase the diversity and randomness. The advantages and disadvantages of this feature remain to be investigated.

2.2.2. MTD strategy based on dynamic platform technology

Dynamic platform technology complicates attacks by dynamically changing the characteristics of a computing platform. In other words, it takes the execution environment as a movement parameter, including the running application and configuration information. Application features include hardware and operating system attributes [36]. Similar to software transformation techniques that typically produce multiple software variants for dynamic switching, this type of technology also typically has multiple instances. This part mainly focuses on the following aspects.

Thompson et al. [40] propose a Multiple Operating System Rotation Environment (MORE) that provides higher security through platform diversity and frequent operating system rotation. This environment makes it difficult for an attacker to detect the vulnerability of the operating system and launch an attack, and it only needs to be based on existing technology and is easy to deploy. However, this environment currently only achieves operating system diversity, and it does not help if the attacker chooses the platform on which the operating system resides.

Software-Defined Network (SDN)-based frequency-minimal MTD method [41] provides a heterogeneous virtual machine pool for cloud service migration. The goal of this approach is to protect critical cloud applications against Loss of Availability (LOA) attacks, such as DoS attacks. In addition, this method is expected to reduce extra resource waste by minimizing the transplantation frequency, which is related to the statistical pattern and probability of DoS attacks. Therefore, the consistency between the attack model is considered by the practical author of the method and the actual attack behavior.

Fulp et al. [42] proposed a framework for using evolutionary techniques to create multiple functionally equivalent but more secure configurations based on existing configurations. This method makes the system configuration presented to the attacker change constantly, which can effectively confuse the attacker and increase the cost of a successful attack. But the deployment costs of this approach are high.

Peng et al. [25] proposed a service deployment strategy under a cloud platform, hoping to make the provided cloud service resistant to attack for as long as possible. The strategy has a risk perception mechanism, which is helpful in improving the effectiveness of the moving target defense mechanism. When cloud services are intensive and/or attackers are highly attackable, it is appropriate to deploy this strategy. When the cloud service is a sparse service and the attacker's attack is weak, the service with mobile target defense does not provide better resilience than the static service.

2.2.3. MTD strategy based on network address shuffling technology

Network address shuffling technology takes the network address as a moving parameter, and the shuffling address can make the address carried by the message in the network random and change with time, thus confusing the attacker. Even if the message is intercepted, the address information is only valid for a short period of time [43].

MT6D [44] is a network layer moving target defense method realized under IPv6, which dynamically rotates the network layer and transports

layer address of the source and destination of both sides of communication to combat eavesdropping, attack and host tracking of specific targets. Network Address Space Randomization (NASR) [45] protects against worm attacks by adjusting the change frequency of node IP addresses at the environment of dynamic network address allocation. This mechanism is transparent to the user and does not require any changes to the protocol or the client. However, it requires changes to the Dynamic Host Configuration Protocol (DHCP) server and the mechanism should be deployed on the network with dynamic addresses, so deployment costs are high [46].

Self-shielding Dynamic Network Architecture (SDNA) [47] is creative in the existing network technology, the hypervisor technology, and authentication technology based on Common Access Card (CAC) and IPv6 technology. It changes the combination of the network in the form of complementary in order to improve the overall security architecture. The technology is transparent to the operating system and compatible with existing network infrastructure and security technologies, which ensures that the operating system cannot be accessed without user-specific authentication and thus limits an attacker's ability to collect and spread information across the network. But the request message before reaching the final destination at least goes through an intermediate node, and the way to propagate from the source node to the destination node is to gradually establish a secure channel in the middle while multiple key exchanges and authentication are required at a cost in the establishment process, resulting in high deployment cost and complexity.

2.3. MTD evaluation method

System evaluation is an important part of system design, and it is no exception in the field of mobile target defense. The main goal of the evaluation is to evaluate and compare the effectiveness of existing defense mechanisms, and study how to improve their effectiveness and provide certain references and guidance for the subsequent design of mobile target defense. In this section, we divide the existing models and methods for evaluating MTD strategies into four categories [48,49].

2.3.1. Evaluation method based on simulation experiments

Zhuang et al. [50] used NeSSimulator2 to create a simulation test bench to test the effectiveness of the proposed MTD system design framework. With the help of the MTD system, SDNA is used as the security policy enforcement unit for each virtual machine. In the experiment, VM refresh is used as the MTD technology, and the attack is guided by a conservative attack graph. VM replacement means that at each simulation interval, the configuration manager randomly selects a role, shuts down the virtual machine that plays the role on one host, and randomly restarts the virtual machine that plays the role on another host, giving the new virtual machine a VM ID and an IP address. The purpose of this experiment is to explore the effect of random change of some properties of the system on reducing the success rate of attacks. However, this work is based on the MTD scheme designed by itself, so it is not universal.

2.3.2. Evaluation method based on theoretical analysis

Han et al. [51] proposed using network propagation dynamics theory to characterize the effectiveness of MTD technology. They first divided existing MTD technologies into three categories: network-based MTD, mainframe-based MTD, and ancillary device-based MTD. Each of these MTD technologies corresponds to the dynamic network transmission model with dynamic attack and defense structure, the dynamic network transmission model with dynamic parameters, and the dynamic network transmission model with both dynamic structure and parameters.

2.3.3. Evaluation method based on model analysis

Evans et al. [52] proposed an effectiveness analysis model of dynamic diversification defense technology. Since the attacker may use five different defensive strategies for an attack, it also determines the five

Table 2

Summary of the evaluation methods of MTD.

Category	Reference	Method used	Contribution
Based on simulation experiment	[24]	Simulation experiment	The effectiveness of the proposed network MTD system was evaluated, and it was found that changes in frequency would affect the success rate of the attack
Based on theoretical analysis	[56]	Theoretical analysis	A general method based on network propagation dynamics and two evaluation criteria are provided to characterize the performance of MTD
Model-based analysis	[52]	Game model	Consider five attack scenarios to assess the effectiveness and conclude that MTD is not always effective
	[53]	Stochastic Petri Net model	Resources that are exposed to an attacker, as well as network resources that have been compromised and can be used to access the system
Based on the hybrid approach	[54]	Urn model and simulation experiment	The effectiveness of network address shuffling technology is analyzed, and four factors affecting the attack success rate are identified
	[55]	Urn model and simulation experiment	The effectiveness of port hopping technology is analyzed, and four factors affecting the defensive effect are identified
	[17]	Urn model and simulation experiment	The effectiveness and performance of network address shuffling technology and the integration of the two technologies were evaluated and compared

attack scenarios in which MTD is valid and invalid and how the speed of re-diversification affects the success rate of attackers. However, this paper only conducted qualitative analysis and lacked quantitative data support.

Moody et al. [53] used Stochastic Petri Nets (SPN) to model and evaluate Defensive Maneuver Cyber Platform (DMCP), which deployed both mobile target defense and decoy defense. By using SPN for the composition of the platform of each node state and the state of the entire platform system modeling, the authors analyzed the defensive mobile network platform of the balance between safety and operability. The influence of system performance was evaluated and analyzed.

2.3.4. Evaluation method based on hybrid methods

Carroll et al. [54] used the urn model and simulation experiments to analyze the performance of network address shuffling technology. By deploying two extreme transformation strategies, static address and perfect transformation, they found that address translation technology can only provide someone with some protection and a less vulnerable network.

Based on their work, Luo et al. [55] utilized the urn model and simulation experiments to analyze the defense capability of port hopping technology against reconnaissance attacks, considering deployment scenarios and drawing conclusions similar to the literature [54]. Besides, the urn model was also used by Crouse et al. [17] to compare network address shuffling, honeypot, and the effectiveness and performance of their combinations in defense against reconnaissance attacks. Table 2 summarizes the evaluation methods of MTD.

3. Reviewing the architectures, enabling techniques, and methods for mimic defense for cyber security

3.1. The introduction of mimic defense

3.1.1. Definition

Mimic defense is a new type of active defense technology for cyber space that is inspired by mimic in the larger world of nature [57]. The background of this new mimic defense is that the perfect state without loopholes and backdoors would not be realized in the current ecological environment or information system of cyberspace [58]. However, most of the traditional defense means are to constantly close the loopholes and find the backdoor to repair the lag, so how to realize the information system with high security in the information device with vulnerabilities is the key problem to be solved urgently in the network security [59,60]. The main idea is to organize multiple heterogeneous functional equivalences to jointly handle the same external request, and make dynamic scheduling based on negative feedback among multiple redundancies to make up for the static, similar and single security defects in the current network space information system or defense technology [61].

Mimic defense is a kind of endogenous security architecture technology based on generalized robust control. It turns uncertain risks into probabilistic events and resolves them all together. The mimic defense has natural immunity to unknown vulnerabilities and viruses in the architecture, and effective integration with existing passive defense methods can form the ability to resist known or unknown attacks in cyberspace [62,63]. However, the mimic defense does not attempt to solve all security problems in cyberspace once and for all, nor does it expect to build any security protection system independently. It does not exclude any defense system and technical means that have been proved to have a security effect, let alone hinders the acceptance of new security technologies or methods that may emerge in the future [64]. In a word, mimic defense is complementary to the existing network space security defense systems. It has the technology fusion as well as the independent controllability in the product and also affects the information system hardware and software. At the same time, endogenous security will also be a necessary capability of information systems in the future [65,66].

Different from traditional network defense, mimic defense is a new active defense based on mimicry. Mimic computation changes the conventional idea that computer applications adapt to computer structures. It is an adaptation of structures to applications to improve energy efficiency. The popular explanation of mimic defense is that in the face of attack, the network system has the ability constantly to change the structure to make the attacker's attack invalid [48,67]. Since the attack is designed to address the weakness of the previous structure, switching to the new structure will have a greater chance of being immune to the attack. If not, it will continue to switch. It is the robustness of the closed-loop system that greatly improves the system's own resistance.

3.1.2. Characteristics

The first character is the basic characteristic of mimic defense. According to the characteristics of the attack chain that relies on the traditional system architecture and operation mechanism, the multi-dimensional reconstruction technology and dynamic as well as randomized security mechanism are combined to disrupt the attack chain to increase the difficulty of attack and realize active defense.

The components containing poisonous bacteria are tolerated, and the software, as well as the hardware components containing poisonous bacteria, are allowed to be used to a certain extent so that the security risks can be controlled [68].

Kernel security risk only depends on the randomness of non-closed dynamic parameters such as current resource state, quality of service, operating efficiency, exception, traffic characteristics and time benchmark of the system, which is the basis of kernel security.

The fusion defense system can enlarge the effectiveness of the security defense measures, and organically integrate the existing security defense

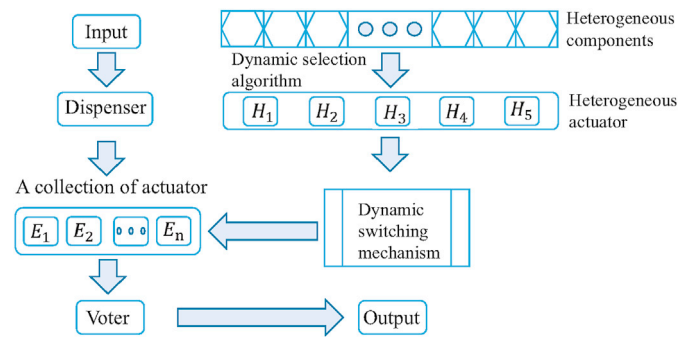


Fig. 3. Typical Dynamic Heterogeneous Redundancy (DHR) of the system architecture.

means. Besides, the system constitutes the active and passive fusion defense system to double the effectiveness of various security technologies through the deep combination of the mimicry mechanism [69,70].

The second is security at the expense of simplicity. The inherent redundancy of the mimic computing architecture makes the mimicry security defense system have inherent reliability. According to security demand, redundancy, cost and reliability index are greatly improved.

Redundant resource application mode based on resource redundancy configuration brings a new effect of the redundant operation. This effect can form special operating mechanisms such as symbiosis cooperation, equivalent multiple variants, and heterogeneous environment migration, which provide innovative methods for the operation of virus tolerance and invasion tolerance, and timely detection, suppression, blocking and removal of trojans and viruses [71,72].

3.1.3. Pivotal techniques

The basic principle of the mimicry defense model is typical Dynamic Heterogeneous Redundancy (DHR) of the system architecture, as shown in Fig. 3.

The basic structure of a computer system is to input data from an input device, process them through an arithmetic unit in a computer, and display the results on an output device, i.e., Input-Process-Output (IPO). In typical DHR architectures, handling ways have been changed. In the pre-treatment stage, the input is copied N times and distributed to the N heterogeneous executive body. Each heterogeneous executive body needs to complete the same function, but all of them are independent of each other. A heterogeneous executive body will send the execution results to the voter after processing has been completed, and the voter can cope with the multiple outputs. Finally, the voter will get the correct output and send it to the user [73].

The basis for dynamic heterogeneous redundancy to play a role is heterogeneity. The larger the property gap between the heterogeneous executor is $\{S_1, S_2, \dots, S_n\}$, the less likely it is to have the same vulnerability. Otherwise, DHR will become formal heterogeneity and isomorphism, and the protection capability will be greatly reduced. The finer the division of heterogeneous components is, the more attributes there will be, the greater the property gap of the heterogeneous executor will

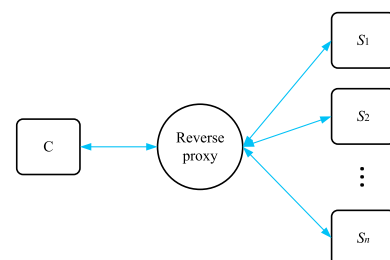


Fig. 4. The structure of the proxy server.

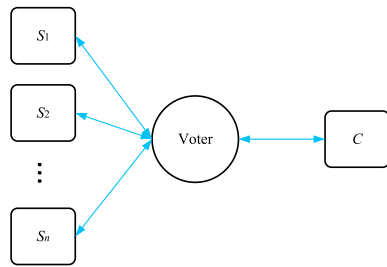


Fig. 5. The structure of the voting device.

be, and the better the defense capability will be [74]. Isomers can be obtained in various ways. One is to obtain them directly by using software diversity. Different software implementations have different vulnerabilities, which is a natural heterogeneity. The other is artificial isomerization, which uses keyword tagging, file tagging, directory randomization, and other methods to artificially make data isomerization [75]. The existence of natural heterogeneity is dependent on the heterogeneity of the software itself. If the difference between the two software packages is small, there is a homologous problem. Artificial isomerization is generated by user customization, so isomerization effect is better and security is higher.

The dispenser is used as early as in the traditional Web software applications, and Nginx has been used as the reverse proxy server. For example, the reverse proxy server does not deal directly with the user request, just as the request of the recipient. It transmits the request to the backend server and transmits the results of the backend server to the users. Its deployment structure is shown in Fig. 4. It can be seen from the architecture diagram that the reverse proxy server plays the role of distributor, but the difference is that the distributor needs to send the request to n backend servers, while the reverse proxy only needs to forward to one of the [52].

As shown in Fig. 5, the voting device votes on the outputs of n identical but independent actuators according to the rules, shielding the errors of fault units to ensure the correct output of the system. The most commonly voting model is that n selects the k -voting model. As long as there is at least k executive body working properly, the entire system operation is normal, in which $n = 3$, and $k = 2$ is the most commonly used three modes of redundancy architecture. Triple modular redundancy that needs two or more executive bodies to malfunction or to be attacked at the same time will result in output errors.

Based on the unreliable hardware, software system and unreliable executable output, the voting algorithm obtains relatively correct data. Due to this feature, the voting algorithm is widely applied. Some hardware system sensors, through the fusion of multiple sensor data, can obtain relatively correct output, the storage system can also be used to improve data reliability. Some highly reliable system control layers can also be used for target detection, pattern recognition, data checking, etc. The voting algorithm usually has multiple inputs, and the voting machine votes on the inputs according to the agreed consistency condition of the system, and finally gets a relatively correct result. Consistency refers to that n input redundant modules of the voting machine transmit message sequence to the voting machine $\{x_1, x_2, \dots, x_n\}$. Set a threshold value t , if there are x_i and x_j and $x_i - x_j \leq t$, then redundant modules i and j satisfy the consistency.

3.2. Theoretical framework of mimic defense

3.2.1. Analysis and modeling of network attack behavior

There are many methods of network attack modeling, mainly focusing on attack language, attack tree, attack network, state transition diagram, and attack diagram. The mimic defense system abstracts the information system into five layers and extracts the variable elements of

each layer for formal description. Information system of defense against the attackers needs to use 5 layers and its elements for attack purposes. To achieve network attack behavior, the attack process is analyzed to extract system knowledge to build a knowledge map and unify the attacking surface and knowledge flow to set up a network attack chain model.

3.2.2. Theory of mimic transformation

According to the idea of mimic security, the mimic transformation can be defined as follows: $\sigma : \Omega(t_i) \rightarrow \Omega(t_{i+1})$. The domain of the transformation is the set Ω of all the states of the system, and the domain is also Ω . For the transformation method of different elements, corresponding to different transformation, so remember $\sigma = \{\sigma_1, \sigma_2, \dots, \sigma_n\}$, where σ_1 represents the mimic transformation of the first element, and so on. The objectives of the mimic security system can be formalized as:

- **Condition one:** $sr(\Omega(t, \sigma)) \leq a$, where sr represents $\Omega(t)$, and the function is used to compute the storage of resources such as interconnections.
- **Condition two:** $pf(\Omega(t, \sigma)) \leq b$, where pf represents the performance function of $\Omega(t)$ and b represents a constant.

Table 3

Mimicry transformation at different levels in mimicry defense system.

Constituent elements	References	Basic elements	Major mimicry transformations
The network layer	[69]	Address, protocol, port, etc.	Change the IP address of the target information system Change the port of the target system Change the protocol used by the target system A combination of the above basic transformations
The platform layer	[71]	Operating systems, etc.	Change the operating system Switch heterogeneous devices Change the virtual machine instance Change the storage system. The superposition of the above transformations in various forms
Environment layer	[73]	Instruction set, etc.	Instruction set randomization Address space randomization The superposition of the above two forms
Software layer	[74]	Heterogeneous variant, etc.	Switch software variant Change the sequence and form of execution instructions Dynamic storage resource allocation scheme The superposition of the above transformations in various forms
Data layer	[75]	The distribution of data.	Change the form of the data Change the syntax of the data Change the encoding of the data The superposition of the above transformations in various forms

Table 4
Effectiveness analysis of the mimic defense mechanism.

Mechanism	Information	Access			Extract		Theft or destruction
		PS	BFA	PT	CP	EV	
Heterogeneous redundancy				✓		✓	
Single line connection	✓					✓	✓
Fragmentation and fragmentation	✓					✓	✓
I/O agent	✓						✓
Stochastic dynamic	✓	✓	✓		✓	✓	✓

Under condition one or condition two, the appropriate mimic transformation σ is selected to realize the function $Max(sp_t|\sigma)$.

Mimic transformation enables the mimic security system with the characteristics of randomness, dynamics and diversity, and effectively improves the certainty, statics and similarity of the traditional system, thereby improving the security of the whole information system. Table 3 lists the main elements and some transformations designed for the five-tier architecture of the information system.

3.2.3. Construction method of mimic security system

The construction method of mimic security system mainly includes the situation awareness method, mimicry method, and cooperative method [76,77]. The studies of the mimic methods include randomization, diversification and dynamic simulation of mimicry security systems which provide the realization and combined application of dynamic randomization mechanism, input-output proxy mechanism, heterogeneous redundancy mechanism, slicing and fragmentation mechanism, single-wire connection mechanism and spoofing mechanism [78].

3.2.4. Effectiveness analysis

In general, to complete the attack task, the attack process can be decomposed into several steps, and different mimicry mechanisms may work in different steps. The following is the effectiveness analysis of the mimicry defense mechanism based on the network attack chain model, as shown in Table 4.

Under a heterogeneous redundancy mechanism: the probability of successful access and right lifting attacks is reduced because these two steps generally rely on vulnerabilities and backdoors, and the redundancy mechanism can effectively defend against attacks relying on vulnerabilities and backdoors.

Single-wire connection mechanism: under single-wire connection, due to different permissions of sensitive paths or key links, information collected and stolen will be incomplete under different permissions, thus reducing the probability of success.

Slicing and fragmentation mechanism: since file and system information is stored and transmitted by slicing and multi-path, respectively, the probability of information obtained by attackers can be effectively reduced, so it is effective for information collection and information theft.

Under the probability model of attack chain: under different mimic mechanisms, each attack chain can reduce the probability of a successful attack of some steps to reduce the probability of a successful attack of the whole attack chain. Therefore, the attack success rate of mimic security systems is lower than that of traditional information systems.

4. Implementation and evaluation of cyber defense

4.1. Implementation and evaluation of network defense in Cyber Physical Power System

The information side and the physical side of the Cyber-Physical

Power System (CPPS) have close and complex interdependence, which is divided into direct dependence and indirect dependence according to the influence mode. Direct dependence refers to the fault on one side that leads directly to the incorrect manipulation or shutdown of the components on the other side [79]. Meanwhile, Indirect dependency is that a fault on one side does not affect the other immediately, but impinges on the ability of its components or system to withstand other disturbances. Hence, the research of CPPS should take into account the physical coupling relationship of the power network security. In view of the information field, the information side attack path, the physical side target and the manipulation method, collectively referred to as the attack vectors, are fused and analyzed in depth [80,81].

Different from the traditional cyber-attacks on information domains, the target of the network attack on CPPS is the power industry control system [82]. The aim of attackers is not only to obtain economic benefits by stealing and manipulating information, but also to damage the stable operation of the power physical system, causing large-scale power supply interruption and other actual physical effects. Therefore, the study of CPPS network attack and defense should include not only the analysis of network attack and the protection of security on the information side, but also the weakening and recovery of function as the ultimate goal on the physical side [83, 84]. It is necessary to consider the support and influence of the information side services on physical side functions, explore the mechanism of attack propagation and effect on the information side and physical side based on direct and indirect dependence of information physics, and form a comprehensive network security protection theory from the aspects of modeling, evaluation, detection and protection [85,86].

4.1.1. Network attack model

Network attack against CPPS refers to tracking the communication system and control system behavior without permission to destroy or reduce power CPS functions and attack the system itself or resources by utilizing vulnerabilities and security defects in power information and communication network [87]. Network attacks can be divided into integrity, availability and security attack.

The most striking feature of network attacks is that attack methods and means vary widely and rapidly. The direct and indirect dependence between the information side and the physical side makes the subtle changes in the attack actions which may trigger different CPPS responses.

The complex and changeable characteristics of attack steps require the adaptability of network attack modeling methods. Meanwhile, the adaptability of network attack modeling methods needs to be higher.

The study improves the network attack modeling method to adapt to the characteristics of CPPS in the field of information and communication, and uses the related functional interface of the CPPS component model to reduce the complexity of process modeling [88].

The information physics hybrid modeling method focuses on the real-time interaction and coupling characteristics of the information side and physical side in CPPS, which considers the corresponding relationship between the attack process and the physical side response. The hybrid modeling method grasps the overall state change of CPPS in the whole process of attack, which reflects the interactive process of attack and defense at a multi-space-time scale and lays a foundation for attack detection and protection [89,90].

The modeling method of human intention incorporates subjective volition into the attack model. In the game, the players of attack and defense follow the principle of the highest to conduct attack and defense [91]. In the original human factor modeling, the influence model of the environment, psychology, workload and other factors is used to model the human decision-making process in the CPPS attack and defense.

4.1.2. Security assessment of CPPS network attacks

Considering the threat of network attack, CPPS security expands the connotation of information security and control security based on the traditional connotation of power grid security and stability. The physical side of CPPS is integrated into this information security assessment

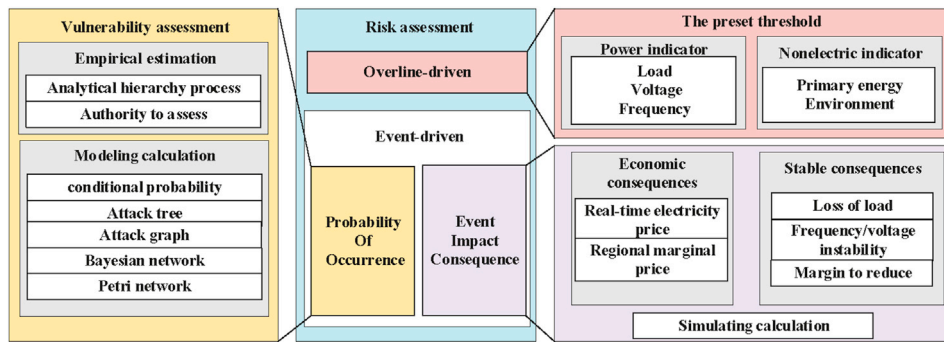


Fig. 6. Relationships between the security assessments for CPPS.

system, which mainly includes CPPS vulnerability assessment and risk assessment [92]. Vulnerability refers to the vulnerability of a powerful information system or secondary system that can be exploited or triggered by a threatening source [93]. The vulnerability assessment refers to assessing the possibility of exploitation of the vulnerability points mentioned above. CPPS security risk refers to the potential impact on CPPS functions caused by network attack threats. The risk assessment refers to the assessment of the expected impact degree of CPPS under threat [94]. The risk analysis is based on vulnerability analysis, which integrates vulnerability assessment and physical consequence assessment. The relationship between vulnerability assessment and risk assessment in the security assessment of CPPS is shown in Fig. 6.

Risk assessment includes the out-of-limit driven method with the preset threshold as the important reference standard and the event-driven method with the potential event as the evaluation object. The event-driven method can be divided into two sub-parts: the probability assessment of event occurrence and the effect assessment of event impact. The probability assessment of events corresponds to the results of vulnerability assessment, which includes the empirical estimation and modeling calculation [95]. The event consequences mainly refer to physical consequences, including economic consequences and stability

consequences. In order to implement the vulnerability assessment, a mathematical model is needed to abstract and quantify the attack process, and a modeling analysis is necessary with the corresponding knowledge of the network attack process [96,97]. Meanwhile, the simulation analysis simulates the attack process on the simulation platform, which restores the whole attack process to a greater extent.

4.1.3. Defense detection of CPPS

The process of defense against CPPS network attacks can be divided into two parts: detection and protection [98]. The purpose of detection is to find the attack behavior suffered by the system in real time, and the purpose of protection is to protect the system from the harm of attack behavior or reduce the harm consequence.

In essence, the defense detection of attack is to judge whether there are abnormal events in the system. The existing anomaly detection methods can be divided into deviation-based detection and feature-based detection according to the identification basis, as shown in Fig. 7. The deviation-based approach usually selects one or more variables that are strongly associated with the attack based on the target system of the defense. An attack is considered to occur when the values of these variables are detected to be too far from the normal range during operation

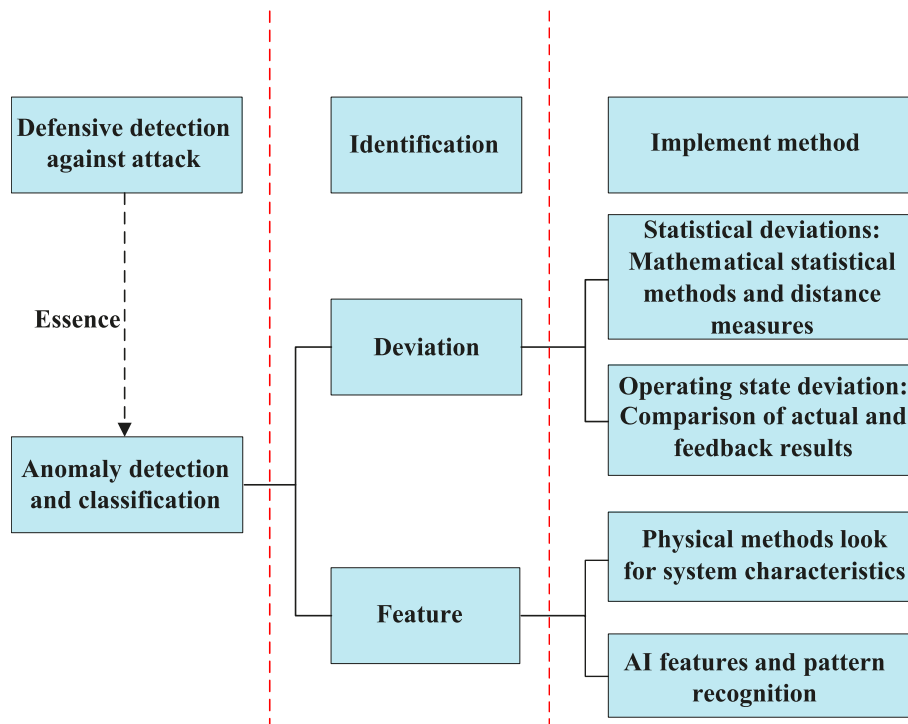


Fig. 7. Attack detection for CPPS.

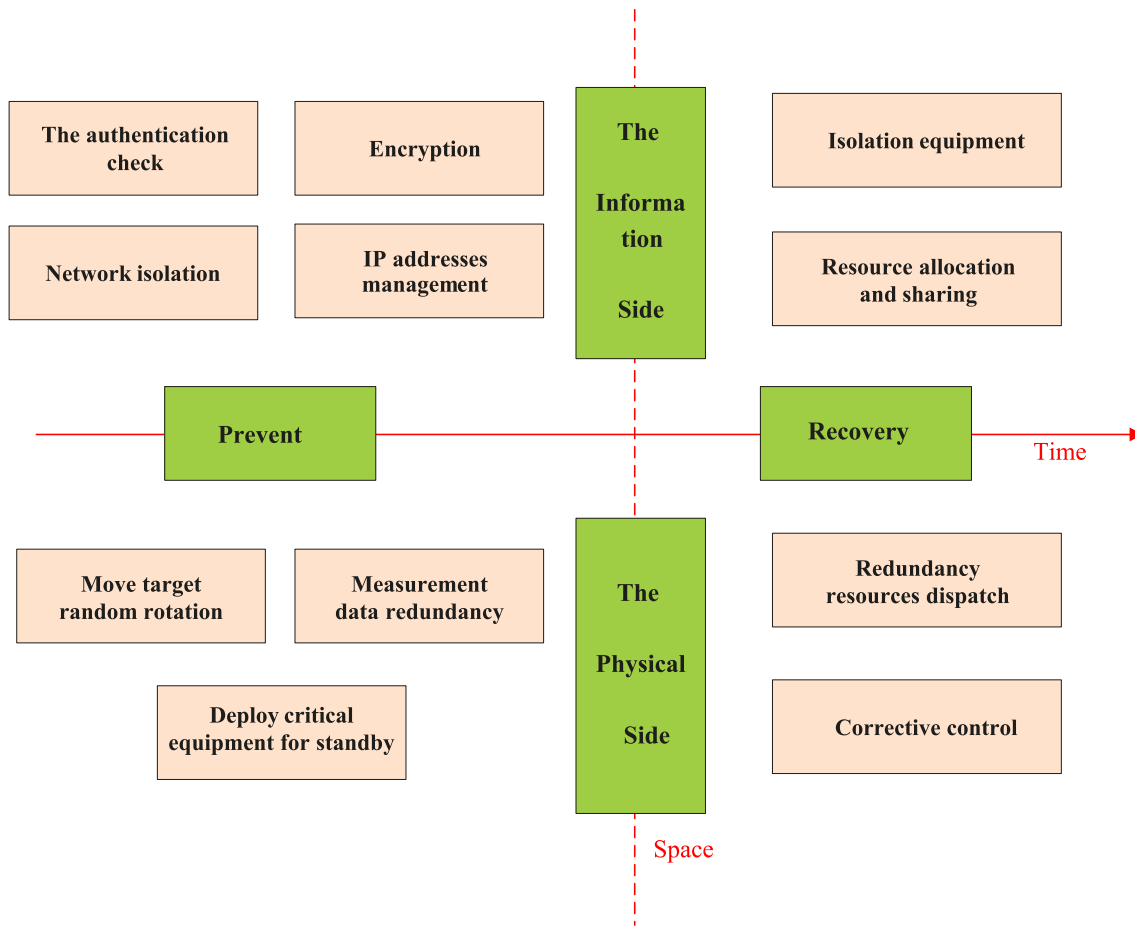


Fig. 8. Attack defense and protection means for CPPS.

[99,100]. The feature-based detection method, through physical mechanism analysis or artificial intelligence method, extracts the features of the system during normal operation and attack, and determines whether an attack occurs by comparing the features in the detection.

4.1.4. Defense protection of CPS

The defensive protection objectives of power CPPS cover two aspects: information security and stable operation of the physical side. The

realization of the protection is through the cooperation of the information side and the physical side defense protection methods as well as the defense methods on both sides, so as to give play to the defense protection capability of the internal security of the one side, and help to maintain the security of the other side through the coupling and correlation nature [101,102]. The ideas of defense and protection are summarized in terms of protection and time scale, as shown in Fig. 8.

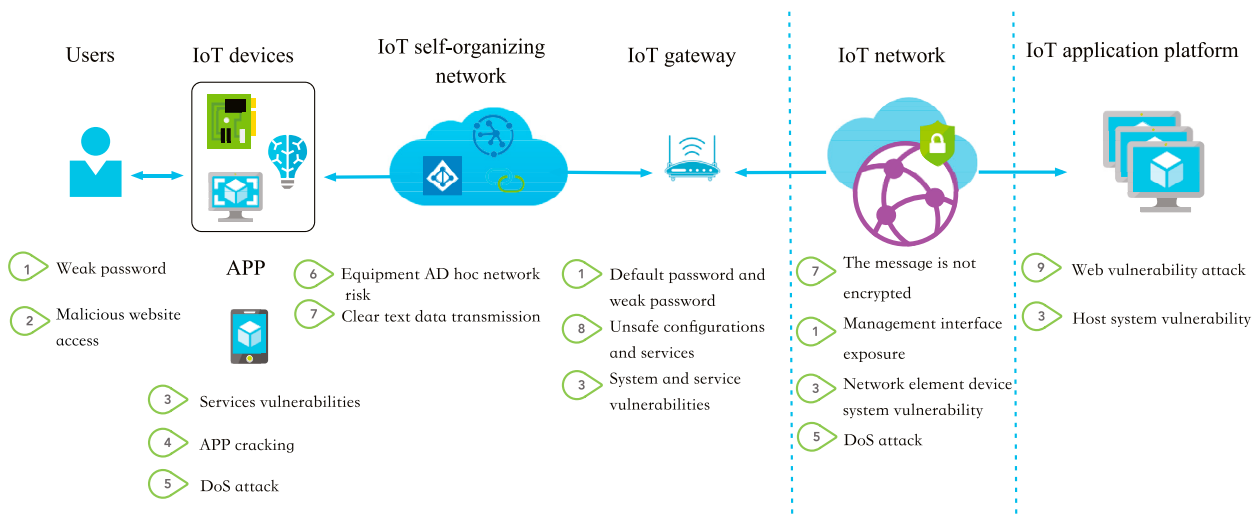


Fig. 9. Key security issues in IoT.

4.2. Network attack and network defense under the Internet of Things

IoT is an extension of the Internet. In addition to inheriting the security problems of the traditional Internet, it also faces the unique problems of IoT security. IoT security system is based on the hierarchical model (including perception layer, network layer, and application layer). The security problems of the Internet of Things are mainly divided into the security protection capability and access security of the terminal devices in the perception layer, the data transmission encryption problem in the network layer, the system security protection of the processing application layer and the important and sensitive data protection of users. These existing problems are often exploited by criminals to attack nodes in the IoT. In order to avoid the security of the situation with great potential, reasonable and effective network defense strategies must be taken. Fig. 9 shows the current security issues in IoT.

4.2.1. Network attack based on IoT environment

Wireless sensor networks are vulnerable to being maliciously attacked because of their characteristics of large-scale distribution, limited node resources and easy capture. Generally speaking, wireless sensor network attacks are divided into internal and external attacks. In defense against external attacks, IoT systems only need to encrypt sent data, decrypt received data again, and initiate integrity checks many times without regularity. The intra-network attack of a wireless sensor is a deliberate attack. This attack pattern is usually a deep security information attack launched after the attacker breaks through the first layer of external security defense, such as wormhole attack, data tampering or Sybil attack. Due to internal attacks, most attackers have disguised their legal identity, and their attacks are highly covert and not easy to be discovered. Such malicious attacks will not only cause data loss and network information chaos but also lead to a trust breakdown of the network node mechanism.

4.2.2. Network defense technology based on IoT environment

As IoT becomes increasingly popular, there are numerous application fields related to IoT that involve every aspect of people's daily life.

Table 5
Summary of the network defense technology.

Cyber defense technology	The corresponding type of attack	Related work	Highlights
Flexible ransomware protection	Ransomware	[103–106]	1. Optimized feature library 2. Isolation of suspicious files and deployment of malicious security code
Flexible defense against botnets	Botnet	[107–110]	1. Abnormal traffic analysis of users and entities 2. Isolate remote code and default credentials
IoT terminal protection	Terminal device attack	[111, 112]	1. Pay attention to equipment safety protection level 2. Minimize new vulnerabilities in software
Safety management and multi-level safety system	Versatile combination attack	[113–116]	1. Reasonable integration of existing institutions, systems, and safety modules 2. Realize the fusion of safety protection 3. Multiple levels of three-dimensional security protection

Therefore, it is particularly important to strengthen the security protection of the IoT and improve it as a whole. Each terminal should take advantage of security technology for effective protection. In the face of network attacks, all nodes should timely and effectively defend to create a healthy and safe network environment and provide people with quality security services.

Flexible ransomware protection: IoT users should pay more attention to ransomware protection and avoid opening suspicious files easily. Malicious code protection software needs to be properly deployed to achieve centralized maintenance. IoT systems should strengthen the protection of the system by opening software protection functions and updating and optimizing the feature library in time. Therefore, data protection technology related to ransomware development is a key step to meeting the current security of the IoT. Meanwhile, intrusion detection technology should be applied flexibly. Once the network defense system finds virus intrusion information, the corresponding nodes immediately get alarmed and begin to work to avoid the virus extortion intrusion and prevent serious losses.

Flexible defense against botnets: Analyzing abnormal traffic by user and entity behavior is the most effective botnet defense technology, which can block abnormal traffic in time. IoT device vendors should avoid using insecure codes and default credentials, consolidate and upgrade devices, and turn off unwanted services. Reasonable configuration of the application of the firewall strategy effectively ensures computer security and plays a role in isolation, preventing unknown security risks, protecting the security of data information, and avoiding the existence of risk information and viruses in the network. At the same time, a combination of firewall technology and intrusion detection technology is an important direction of current defense technology. Through an effective combination of the two, the overall optimization of the current function is promoted. Firewall technology can be effectively applied to protect abnormal data in the outside world to give full play to the role of security mechanism and security protection. Root intrusion detection technology uses the same principle to detect intrusion in the network. Therefore, the two need to be effectively combined to give full play to the comprehensive advantages to meet the current needs.

IoT terminal protection: IoT industry manufacturers should adopt the best safety protection design scheme according to the characteristics of their own equipment and improve the safety protection level of their own equipment. Timely updates need to be provided to minimize new vulnerabilities in the software. In addition, communication mechanisms need to be encrypted to prevent data from being transmitted in clear text over the network. The intrusion detection and protection mechanism, which monitors the malicious intrusion in real-time and alarm in time, should also be established. The IoT system regularly introduces the third party to conduct security testing and evaluation for the IoT products, providing reliable and authoritative security guarantees.

Safety management and multi-level safety system: In the current IoT environment, all aspects of the system should actively establish a sound management system, such as the existing organization, rules, regulations, information security and other related functional equipment for reasonable integration, to achieve the integration of security protection and promote the overall security of the system to improve. The strengthening of information protection is the key step to creating a great network environment for people. Safety management should start with technology and strategy. Analysis needs to be performed in-depth to optimize the IoT as a whole to give full play to their respective roles and ensure network system security. In fact, the current network security system has many levels, such as common structure level, data security level, transmission security level and permission level, so it should be improved according to the actual needs of the current, and according to different security requirements to provide the corresponding security algorithm and system. Therefore, through the improvement of multiple three-dimensional security protection levels, the management security system is effectively constructed to achieve targeted management, give full play to the advantages of multiple technologies, and ensure the

network system in the process of operation with security.

We give a summary of the network defense technology in Table 5.

4.2.3. Defense strategy for IoT under evolutionary game

Because the interaction between normal nodes and malicious nodes in the IoT has the characteristics of attack and defense, game theory has been widely applied to solve the security problem of the IoT. However, at present, most of the game models are built on the basis that the nodes of the IoT are in a completely ideal state. The assumption that both sides always adopt optimal strategies is not consistent with the characteristics of the real IoT. In reality, it is difficult for IoT nodes to grasp all the network information, and defense measures may not always be optimal. The evolution game theory is used to analyze the security state change of IoT, which does not require participants to master complete information and conform to the characteristics of IoT nodes. The evolutionary game theory combines game theory with a dynamic evolution process and can analyze the stability of incomplete information evolution by the dynamic system method. In the IoT environment with malicious nodes, to minimize their own risks, network nodes constantly adjust their attack and defense strategies through learning and imitation.

5. Open challenges

According to the comprehensive discussions above on existing efforts, the key open challenges and future research directions are articulated for dynamic defense for cyber security.

5.1. Vulnerability problem

Dynamic defense for cyber security resists attackers by diverting the attack surface. However, system vulnerabilities still exist. Defenses randomize the moving targets such as software, but if the software of vulnerability has not been fundamentally solved, the attacker can still dig through the leaks and buffer overflow vulnerabilities to specific targets. Only with the software after randomization, different users of the binary code are different, and therefore it cannot be used for other goals in the same way to carry out attacks [117]. Another example is instruction set randomization. Although it prevents attackers from inserting binary instructions into the target program to execute the attack successfully, the vulnerability of the target program has not been eliminated, and the well-designed worms and viruses can still break through the defense line of instruction set randomization [40].

5.2. Integration with existing techniques

Existing dynamic defenses for cyber security, such as firewall, intrusion detection system, and anti-virus systems, are deployed in the network. The network topology and configuration are relatively fixed, while the defense of the moving target will change the existing network configuration. Therefore, the network availability may be reduced, and the existing network security defense technology may be interfered with. Mobile target defense technology must be implemented on the basis of not affecting the existing network operation and must adapt to the existing network infrastructure, network services and network protocols. With the deepening of the research, the dynamic defenses for cybersecurity techniques will be better integrated with the existing network security protection technology and be better embedded in the existing network [21,118].

5.3. Systematic development

At present, abundant researchers propose various attack surface transfer schemes based on the moving targets defense idea. However, the schemes have not formed a system, and the overlapping use of different moving targets defense techniques may lead to conflicts. As a result, the analysis of the influence on the moving target defense technology system

or network attributes and the judgment of the stack using different moving target defense techniques to form a dynamic defense for cyber security system is an important work in the future [119].

5.4. Integration with emerging techniques

Dynamic defense for cyber security tends to change network configuration, which results in the loss of availability. The IP address change interferes with the attacker's scanning and intrusion, but may cause the failure of the entire network communication. In addition, the new software to define network SDN fundamentally changes the network structure, which makes the central controller have the ability of global regulation in the network. Therefore, based on the SDN technique, the change of IP makes the dynamic defense for cyber security technique minimize the impact of the entire network [25].

6. Conclusion

With the rapid development of various computing paradigms, information resources are widely shared and fully utilized. Consequently, cyber security problems are aggravated. To cope with this challenge, moving target defense and mimic defense are investigated to improve the defense effect. Furthermore, improving dynamic defense system construction has important theoretical guidance and practical significance for improving network active defense capability.

In this paper, a comprehensive survey of recent research on dynamic defense in cyber security is conducted. Technically, the background and motivation for the dynamic defense in cyber security are first reviewed. Then, an overview of the frameworks, architectures and emerging key techniques for cyber security is provided. Afterwards, the implementation and evaluation of dynamic defense are discussed. Finally, the open challenges and future research directions on dynamic defense in cyber security are investigated. We hope that the survey is able to elicit further discussions and research on dynamic defense in cyber security.

Declaration of competing interest

The authors declare that they have no competing interests.

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